MTD315

Timed Online Assignment – January Semester 2022

Computer Interactive Graphics

Saturday, 21 May 2022 **4:00 pm – 6:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This Timed Online Assignment (TOA) contains **FOUR (4)** questions and comprises **FIVE (5)** pages (including cover page).

2. You must answer **ALL** questions.

3. If you have any queries about a question, or believe there is an error in the question, briefly explain your understanding and assumptions about that question before attempting it.

4. You MUST **submit your answers via Canvas (similar to TMA submission) at the end time of this TOA** (as stated on this cover page). The 15 minutes grace period as shown on Canvas is strictly meant for technical issues encountered during submission. Thereafter, you will not be able to submit your answers and you will be considered as having withdrawn from the course. **No appeal will be allowed.**

5. You are to include the following particulars in your submission: TOA Course Code, your Full Name, and SUSS PI Number. Name your submission file as TOACourseCode_FullName_StudentPI. Use underscore and NOT Space. *Example: TOAXYZ_RaphaelLee_T1923161 (omit D/O, S/O).*

6. Your submission should consist of only one file and **must not exceed 500MB** in size. The file must be a Microsoft Word file saved in .doc or .docx format. **All answers are to be typed** except for codes, formulas, calculation workings, flowcharts and graphs which can be handwritten. Handwritten codes, formulas, calculation workings, flowcharts and graphs may be scanned or photographed and embedded in the Word file provided it does not exceed the file size limit of 500MB. Please review your submitted TOA work after uploading it into Canvas to make sure that any handwritten work can be seen clearly.

7. To prevent plagiarism and collusion, your submission will be reviewed by Turnitin. The Turnitin report will only be made available to the marker and you will not be able to view it.

8. The University takes plagiarism and collusion seriously, and your Turnitin report will be examined thoroughly as part of the marking process.

Answer all questions. (Total 100 marks)

Question 1

Employ the use of attributes to describe how an object should be rendered. Each type of objects would have its own set of available attributes.

(a) Provide examples of attributes that are available to each of the followings:

(i) Primitives such as points, line segments and polygons

(10 marks)

(ii) Stroke text

(5 marks)

(b) Illustrate how attributes such as colours can be bound to a cube.
Note that a cube has six faces and each face is made up of two triangles.

(10 marks)

Question 2

(a) Appraise and describe briefly each of the following two approaches for creating curved objects and surfaces.

(i) Approximation using primitives

(8 marks)

(ii) Mathematical definitions

(8 marks)

(b) Analyse the code for the vertex shader and the fragment shader used in the gasket program:

```
in vec4 vPosition;
void main() {
   gl_Position = vPosition;
}
```

```
void main() {
  gl_FragColor = vec4( 1.0, 0.0, 0.0, 1.0 );
}
```

Using the above codes as examples, explain the purpose of the vertex and fragment shaders.

(9 marks)

Question 3

(a) Two-dimensional viewing may be considered as taking a rectangular area of our two-dimensional world and transferring its contents to the display.

(i) Describe briefly the use of viewing rectangles and object clipping in two-dimensional viewing.

(8 marks)

(ii) Draw a diagram to show objects before clipping and another to show image after clipping.

(6 marks)

(b) Appraise the use of matrices to represent transformation such as rotation, translation and scaling.

(i) Write the matrices that represent the rotation of an object about the x-axis by an angle α, followed by a rotation about the y-axis by an angle β.

(7 marks)

(ii) Identify **TWO (2)** rigid-body transformations. Support your answer with explanation.

(4 marks)

Question 4

(a) Figure Q4(a) is a model representing computer imaging and viewing of objects.

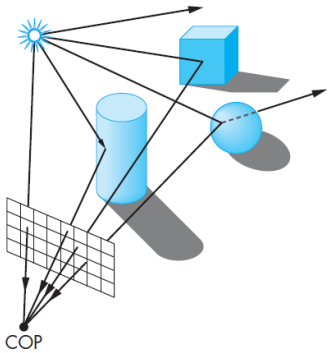

Figure Q4(a): Light, surfaces, and computer imaging

Assemble the elements in the model and describe the role of the project plane.

(10 marks)

(b) Figure Q4(b) shows the effect of local illumination and global illumination.

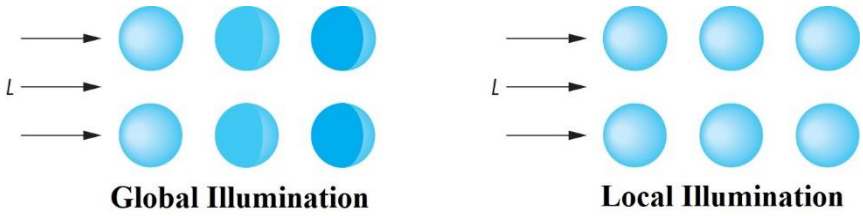

Figure Q4(b): Effect of illumination on shaded spheres

(i) Analyse and discuss how global illumination overcome the limitations of local illumination.

(10 marks)

(ii) The two rendering strategies for handling global illumination are ray tracing and radiosity. State the condition in which each strategy is suitable.

(5 marks)

----- END OF PAPER -----